

1. Name the components of the formed elements in the blood and mention one major function of each of them.

- ☞ Components of formed elements of blood are - RBCs or red blood corpuscles, WBCs or white blood corpuscles and thrombocytes or blood platelets.

The most important function of RBCs is to transport oxygen from lungs to body tissues and CO₂ from body tissues to the lungs. White blood cells provide immunity to the body. Blood platelets play important role in blood clotting.

2. What is the importance of plasma proteins?

- ☞ Plasma proteins constitute about 7 to 8% parts of plasma. These mainly include albumin, globulin, immunoglobulin, prothrombin, fibrinogen, lysozyme and properdin, etc. Prothrombin and fibrinogen are needed for blood clotting.

Albumins are involved in osmotic balance. Immunoglobulins act as antibodies and some other plasma proteins such as lysozyme and properdin destroy bacteria, viruses and toxic substances that may enter into the blood from outside.

3. Match Column I with Column II:

Column I	Column II
(a) Eosinophils	(i) Coagulation
(b) RBC	(ii) Universal Recipient
(c) AB Group	(iii) Resist Infections
(d) Platelets	(iv) Contraction of Heart
(e) Systole	(v) Gas transport

- ☞ (a) – iii (b) – v (c) – ii (d) – i (e) – iv

4. Why do we consider blood as a connective tissue?

- ☞ A connective tissue has 3 components - intercellular matrix, connective tissue cells and fibres. Blood is a connective tissue because it consists of a fluid matrix (plasma), blood cells (RBCs, WBCs & platelets) and fibrous elements in plasma.

5. What is the difference between lymph and blood?

- ☞ Differences between lymph and blood

Blood	Lymph
It consists of plasma, RBCs, WBCs and platelets.	It consists of plasma and WBCs (mostly lymphocytes).
It is red in colour due to the presence of haemoglobin in RBCs.	It is colourless as haemoglobin is absent.
Its plasma has more proteins, calcium and phosphorus.	Its plasma has fewer proteins and less calcium and phosphorus.
Normal amount of CO ₂ and other metabolic wastes.	Much more amount of CO ₂ and other metabolic wastes.
It carries materials towards and away from the tissue. Therefore, it acts as a “vehicle”.	It transfers materials from the blood to the body cells and vice-versa. Therefore, it acts as a “middle man”.

6. What is meant by double circulation? What is its significance?

- ☞ It is the circulation in which blood flows through the heart twice for completing its circuit. Here, oxygenated and deoxygenated bloods never mix up. It includes pulmonary circulation and systemic circulation. Double circulation provides a higher metabolic rate to the body. It also allows the two circulations to have different blood pressures according to the need of the organs they supply.

7. Write the differences between:

- (a) Blood and Lymph
(b) Open and Closed system of circulation
(c) Systole and Diastole
(d) P-wave and T-wave

- ☞ (a) Refer Q. No. 5

(b) **Open circulatory system:** Here, the blood pumped by the heart passes through large vessels into open spaces or cavities called sinuses. E.g. Arthropods and molluscs.

Closed circulatory system: Here, the blood pumped by the heart is always circulated through blood vessels. This is more advantageous as the flow of fluid can be more precisely regulated. E.g. Annelids & chordates.

- (c) Systole is contraction of heart chambers in order to pump out blood. It decreases volume of heart chambers and forces the blood out of it.

Diastole is relaxation of heart chambers to receive blood. It brings heart back to its original size to receive more blood.

- (d) P-wave: It represents the excitation (depolarization) of atria which causes atrial systole.

T-wave: It represents the repolarisation of ventricles.

8. Describe the evolutionary change in the pattern of heart among the vertebrates.

- ☞ In vertebrates, the heart exhibits an evolutionary change in its structure. For example,

In fishes, heart is 2-chambered (an atrium + a ventricle).

In Amphibians, heart is 3-chambered (2 atria + a ventricle).

In Reptiles (except crocodiles), heart is 3-chambered (2 atria + a ventricle), but ventricle is incompletely partitioned.

In Crocodiles, birds & mammals, heart is 4-chambered (2 atria + 2 ventricles).

9. Why do we call our heart myogenic?

- ☞ Normal activities of heart are auto-regulated by nodal tissues. So it is called myogenic heart.

10. Sino-atrial node is called the pacemaker of our heart. Why?

- ☞ Sino-atrial node (SAN) initiates and maintains contraction of heart by generating action potentials (70-75/min). So it is called the pacemaker of heart.

11. What is the significance of atrio-ventricular node and atrio-ventricular bundle in the functioning of heart?

- ☞ From the atrio-ventricular node, a bundle of fibrous atrio-ventricular bundle (AV bundle) passes through atrio-ventricular septa and divides into a right & left branches. Each branch passes through the ventricular walls of its side. In the ventricular wall, it breaks up into minute fibres (Purkinje fibres).

The action potential is conducted to ventricular side by AVN & AV bundle from where bundle of His transmits it through the ventricular musculature. As a result, the ventricles are contracted.

12. Define a cardiac cycle and the cardiac output.

- ☞ The cyclic events occurring in the heart that include atrial systole, ventricular systole and diastole is called cardiac cycle.

Cardiac output is the volume of blood pumped out by each ventricle per minute, i.e. stroke volume x heart rate (70 x 72). It is about 5000 ml (5 litres).

13. Explain heart sounds.

- ☞ During each cardiac cycle, 2 sounds are produced. The first sound (**lub**) is due to the closure of tricuspid and bicuspid valves. The second sound (**dub**) is due to the closure of the semilunar valves.

One heartbeat = a lub + a dub.

14. Draw a standard ECG and explain the different segments in it.

- ☞ An ECG consists of the following waves:

P-wave: Represents the excitation (depolarization) of atria which causes atrial systole.

QRS-complex: Represents depolarization of ventricles (Ventricular systole).

T-wave: Represents the repolarisation of ventricles.

